Development of a Smart Meter with Non-intrusive Load Monitoring Function

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***Abstract*—Currently, demand of precise and refined energy consumption service for residents is growing constantly. Aiming at this situation, according to the principles of International Recommendation 46, an electricity meter for non-intrusive load monitoring based on multicore modularity is designed to improve the level of detailed electrical data analysis. With general function module, metering module and load analysis module embedded separately, intelligent identification for type, turn-on/off time and electricity consumption of household appliances is achieved in triple cores management mode by load data sampling, characteristic quantity extraction, load matching and record upload. At the same time, a remote update scheme relying on electricity information acquisition system is proposed to ensure that load identification algorithm and load signature database could be updated online.**

Keywords—non-intrusive load monitoring, load signature, IR46, multicore modularity, smart meter

I. Introduction

With the continuous improvement of people's living standard, residents' demand for precise and refined energy consumption services is also growing. Household load monitoring analysis, intelligent identification of household appliances types, turn-on/off time and power consumption, supplement of electric appliances' power consumption details, household energy consumption diagnosis, and early warning of abnormal operation of household appliances, etc, are critical requirements for enhancing the electricity service level of the power supply company, for guiding users to use electricity scientifically and rationally, and for promoting energy conservation and emission reduction in general [1-2].

At present, smart meters and the electricity information acquisition system have covered 99% of users within the business scope of the State Grid Company of China, achieving the goal of remote automatic acquisition of residential users' daily electricity data. Compared with residential load analysis scheme such as household smart plugs or load monitoring terminals, the smart meter is able to carry out non-intrusive load monitoring (NILM), and does not need to install various load acquisition and analysis equipment separately, which can avoid the hardware purchase cost, and will not affect normal lives of users [3]. Integrating the NILM function to the smart meter is the preferred way to carry out residential load analysis on a larger scale.

The key of load identification is the load matching algorithms and the load characteristics library [4-5]. In order to obtain the working data sequence of the household appliances, cluster analysis, feature dimension reduction, neural network, fuzzy matching and other algorithms are usually used to analyze current, voltage, power, harmonics and other load signals [6-9]. With the development of technologies, all kinds of intelligent home appliances with new working principles will emerge one after another; load matching algorithm or load characteristics library will need to be updated from time to time to improve the identification

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accuracy and support more electrical appliances types. Therefore, according to the concept of International Recommendation 46 (IR46) international proposal, a smart meter based on the multicore modular solution is designed and developed in this paper, which has load analysis and supports remote upgrade as well. The meter has built-in function module (management core), metering module (metering core) and special chip for load analysis (NILM core). The feasibility and effectiveness of the developed prototype's load analysis function, have been verified by experimental tests of the designated metrological verification agency and live tests of household units.

II. Introduction of NILM Scheme

A. Load Identification Process

Due to different working principles and functionalities, the load characteristics of various kinds of electrical appliances during startup, operation and shutdown are discernibly different [10]. Different appliances have different characteristics. Take the electric kettle and microwave oven as examples. The comparison of power curves is shown in Fig. 1.

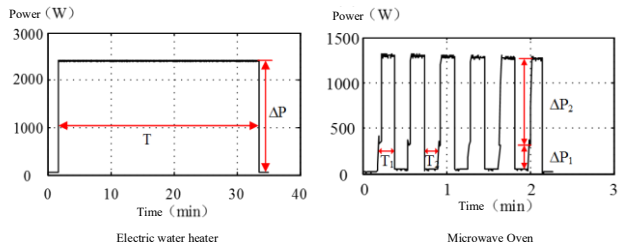

Fig. 1. Comparison of two power curves

In the Fig. 1, it is found that the active power rises quickly to the rated power in a step-like manner when starting the electric kettle, and the microwave oven operates intermittently.

Extracting load characteristics such as voltage, current, active / reactive power, or harmonic waves from the total household load, and then comparing from multiple dimensions with the prior characteristic quantity groups of various electrical appliances obtained with model building, we can decompose the total household load of a certain time period into: the types of operating electrical appliances, the respective start and stop time and the corresponding power consumption. This constitutes the basis for meters to conduct non-intrusive residential load identification.

The overall process of load identification includes four main components: load data collection, data preprocessing, feature extraction and load matching. The first step is the load data collection, which obtains the raw signals of voltages and currents at the users' main household power supply line at the sampling rate of 1.6kHz (32 sampling points per second). The second step is the data preprocessing. The active and reactive powers are calculated through the instantaneous value integration formula, and the harmonic current is calculated through the fast Fourier transform. At the same time, per unit voltage values, data denoising and information compression[11] are employed to provide accurate and reliable data for subsequent analysis. The third step is load feature extraction. In order to improve the accuracy of load type classification, characteristic quantities are usually divided into three categories: steady state, transient and runtime [12]. Finally, a specific algorithm is used to match the extracted load characteristics with the characteristics library containing known electric appliances' characteristic quantity groups to complete the load identification.

B. Load Feature Quantity Extraction

Different types of appliances may have certain load characteristics that are actually identical or similar, but the load characteristic quantity groups composed of multiple feature dimensions is unique, that is, each appliance has its own unique load characteristics.

1) Steady-state characteristic quantity

Among the three types of characteristic quantities, the steady-state characteristic quantities include steady-state active / reactive power, steady-state harmonic current, etc., which are the embodiment of the specific value of the load characteristic quantity. In some occasions, the steady-state characteristic quantity can be directly matched with certain electrical appliances.

The steady-state active power is the average of the instantaneous power over a period, i. e., averaging the integration of the product of the voltage and current sampling values. The steady-state active power P and reactive power Q, used as load characteristic quantities, are:

$$\begin{cases} P = \dfrac{1}{mN} \displaystyle\sum_{k=(n-1)*mN+1}^{nmN} u_k * i_k \\ Q = \dfrac{1}{mN} \sqrt{\displaystyle\sum_{k=(n-1)*mN+1}^{nmN} u_k^2 * \sum_{k=(n-1)*mN+1}^{nmN} i_k^2 - (\sum_{k=(n-1)*mN+1}^{nmN} u_k * i_k)^2} \end{cases} \tag{1}$$

where, m is the number of power frequency cycles contained in the calculation time window of active power series P; N is the number of current and voltage sampling points included in one power frequency cycle; u_k and i_k are sampling values for the discrete voltages and currents.

The steady-state harmonic current, serving as the load characteristic quantity, is obtained from the fast Fourier transform of current sampling values. The relationship between i_a, the current for class a electrical appliances running in steady state, and the other harmonic currents is:

$$\begin{cases} i_a(t) = I_{a1} \cdot cos(\omega t + \theta_{a1}) + \cdots + I_{ak} \cdot cos(k\omega t + \theta_{ak}) + \cdots \\ I_{ak} = \alpha_{ak} \cdot I_{a1} \end{cases} \tag{2}$$

where, I_{a1} is the amplitude of first harmonic; I_{ak} is the amplitude of the k-th degree harmonic wave; α_{ak} is the ratio of I_{ak} and I_{a1}, and can take part in the identification process as a characteristic quantity.

2) Transient characteristic quantity

The transient characteristic quantity includes the step, impulse, gradient, etc., of the active power. The transient characteristic quantity is closely related to the functionality, start and stop of the electrical equipment [13]. Therefore, it is suitable for the identification of complex scenarios such as

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similar single load characteristics or overlapping operations of multiple appliances.

Take the starting of electrical appliances as an example, for non-linear load such as inverter air conditioners, the active power will not immediately enter the steady state at startup, but will produce an active impulse. To measure the extent of the active impulse, the active impulse coefficient k_p was selected as one of the transient characteristic quantities, it is defined as:

$$k_p = \frac{\Delta P_{peak}}{\Delta P} \tag{3}$$

where, ΔP_{peak} is the active impulse peak at load start, and ΔP is the increment relative to pre-start value after the active power becomes stabilized. In particular, for resistive heating loads, $k_p \approx 1$, which means the active power directly enters the steady state in step form.

3) The runtime

For different electrical appliances, even in the same work mode, the operation time is generally different. For example, the electric water heater usually lasts for more than 20 minutes when heating, while the electric kettle does within 5 minutes. On the other hand, for loads undergoing discontinuous operations, each individual running time tends not to be the same. For instance, the washing machine operates in roughly evenly spaced time intervals, while the rice cooker does in non-evenly spaced time intervals. Therefore, the load type and work mode can be distinguished from the time dimension, and the start/stop moments can be clearly defined.

We define T_k as a characteristic quantity representing the runtime of the load under different operating conditions:

$$T_k = T_{ek} - T_{sk} \tag{4}$$

where, T_{ek} is the stop moment of the kth segment of the total running process, T_{sk} is the starting moment of the kth segment. When k =1, it means that the load runs continuously in this segment, otherwise it would mean intermittent operation. When $T_k = T_{k+1}$, it means that the load is operating in evenly spaced time intervals.

C. Load Matching

The premise of load matching is to establish a relatively complete and continuously updated load characteristics library. Load characteristics library is obtained by large numbers of measured waves against actual electrical loads, load characteristics extraction and clustering algorithm for generalization. Usually, the same type of appliances with different brands and power specifications are categorized as the same type of load, while appliances with different working principles are categorized as different types. i.e., the load characteristics of fixed frequency air conditioners and inverter air conditioners are different, and they can be distinguished as two types of loads in the characteristics library. The Fig. 2 is a curve of the parts of the characteristic quantities of the two types of air conditioners in cooling mode.

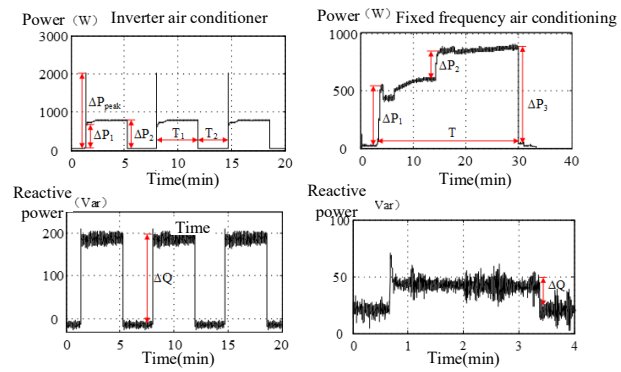

Fig. 2. Load signature database for varied frequency and fixed frequency air conditioners

On top of establishing the load characteristics library, a multi-layer tree classification load matching algorithm is adopted, and the main flow chart is shown in Fig. 4.

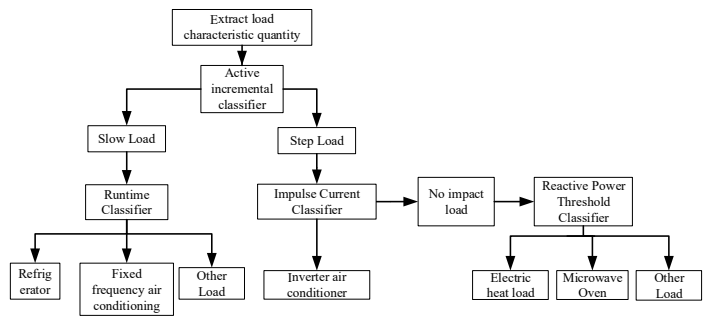

Fig. 3. Main diagram for load matching method based on multilayered classification tree

Through utilizing a multi-feature dimensional classifier, the extracted feature quantities are matched hierarchically. First, the active power increment classifier divides the load into slowly varying load and step load. For slow varying loads, we can distinguish appliances like fixed frequency air conditioners and refrigerators through the runtime classifier; for step loads, we can distinguish inverter air conditioners and other non-impulse loads through the inrush current classifier. Then, for non-impulse loads, through the reactive threshold classifier we can further distinguish electric heating appliances and microwave ovens, etc. At present, the algorithm supports identification of more than 10 categories of home appliances such as fixed frequency / inverter air conditioners, water heaters, electric kettles, microwave ovens, rice cookers, induction cookers, refrigerators.

III. Designation and Test

A. Design of Multicore Modular Electricity Meter

IR46 was issued by International Legal Metrology Organization (OIML). It is suggested that the electronic equipment of the electricity meter should be separated from the components, the metering function is independent from other functions, and the online upgrade of the non-metering part software does not affect the accuracy and stability of the

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metering part. The study of multicore intelligent electricity meter technology based on IR46 standards is in line with the development trend.

According to IR46, only when the metering core is independent of the software and hardware systems of other non-metering cores, the electric energy meter can ensure that the function expansion and remote upgrade of the non-metering part do not affect the metering performance. Therefore, the intelligent electric energy meter based on the multicore modular design can achieve the remote upgrade of the load analysis function.

The hardware system of multicore modular electricity meter with load analysis and remote upgrade consists of three functional chips: management core, metering core and NILM core. The three functional chips are placed inside the electricity meter, each with a microcontrol unit (MCU), which can achieve independent management. The hardware block diagram is shown in Fig. 4.

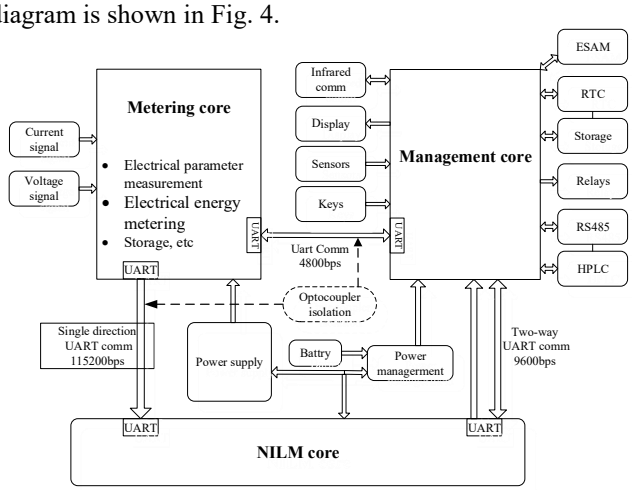

Fig. 4. The hardware block diagram of the multi-cores electricity meter

In Fig.4, in addition to meeting the requirements of multi-core communication, a high-speed optocoupler is added to the communication physical link between the metering core and other cores to achieve electrical isolation to furtherly ensure safety.

B. Load Analysis Function Test

In order to verify the function of the load analysis of the electricity meter, a load analysis electricity meter test platform that imitates the residential electricity use environment was built by real electrical appliances in the laboratory. The test architecture is shown in Fig. 4.

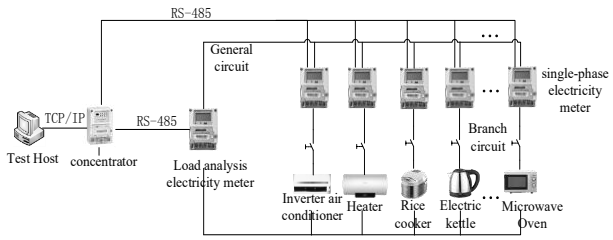

Fig. 5. Architecture of test platform for load identification

The load analysis electricity meter is connected to the main power supply circuit, and different types of electrical appliances connected to each branch circuit. Each branch circuit is additionally connected to a single-phase meter to measure the power load of the branch circuit. All electricity meters upload relevant data to the main test station via the concentrator, and analyze the load decomposition performance. Define the power decomposition accuracy $P_{rec}(k)$ of the k-th electrical appliance as:

$$P_{rec}(k) = 1 - \frac{\sum_{i=1}^{N} |W_{ident,k}(i) - W_{real,k}(i)|}{\sum_{i=1}^{N} W_{real,k}(i)} \tag{5}$$

where, $W_{\text{ident,k}}(i)$ is the electricity consumption (unit: kWh) of the k-th electrical appliance decomposed by the electrical energy meter during the i-th monitoring period, and $W_{\text{ident,k}}(i)$ is the corresponding real electrical energy measured by the brach electricity meter. The monitoring period is 15 minutes. The test results are listed in Table I, and the test results are listed in Table II.

TABLE I. ACCURACY OF POWER DECOMPOSITION IN SINGLE APPLIANCE OPERATING CONDITION

Electrical type	Rated power/W	Prec/%（Single Electrical Appliance）
Fixed frequency air conditioning（wall-mountable）	1120	98.24
Fixed frequency air conditioning（Cabinet）	2350	97.16
Inverter air conditioner（wall-mountable）	1430	98.88
Electric kettle	1800	96.70
Rice cooker	860	97.75
Electromagnetic furnace	2100	95.36
Microwave Oven	1300	97.50
Electric water heater	2000	95.40

TABLE II. ACCURACY OF POWER DECOMPOSITION IN MULTIPLE APPLIANCES COMBINED OPERATING CONDITION

Number of appliances	Switch on electrical type	Identify electrical type	Prec/%
2	Fixed frequency air conditioning（Cabinet） Inverter air conditioner	Fixed frequency air conditioning Inverter air conditioner	85.35 86.20
2	Inverter air conditioner Rice cooker	Inverter air conditioner Rice cooker	97.86 90.85
2	Electric kettle Microwave Oven	Electric kettle Microwave Oven	96.54 97.02
3	Fixed frequency air conditioning（wall-mountable） Inverter air conditioner Microwave Oven	Fixed frequency air conditioning Inverter air conditioner Microwave Oven	81.94 87.29 96.69
3	Rice cooker Electric kettle Electromagnetic furnace	Rice cooker Electric kettle Electromagnetic furnace	94.36 90.45 78.11
4	Inverter air conditioner Rice cooker Electromagnetic furnace	Inverter air conditioner Rice cooker Electromagnetic	85.19 81.67 91.82

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	Microwave Oven	furnace Microwave Oven	93.78
7	Fixed frequency air conditioning（Cabinet） Inverter air conditioner Rice cooker Electric kettle Electromagnetic furnace Microwave Oven Electric water heater	Fixed frequency air conditioning Inverter air conditioner Rice cooker Indeterminate appliances Indeterminate appliances Microwave Oven Electric water heater	77.85 79.19 73.41 80.60 70.14 86.40 91.23

It can be seen that the developed load analysis electricity meter has no load type matching error under single and multi-electrical appliances operation conditions. The power decomposition accuracy of single appliance is 95% - 98%, and that is 70% - 90% under multiple appliances operation condition.

In a complex scenario where more than 3 electrical appliances are in mixed operation, the active power, reactive power, harmonics, etc. are prone to large oscillations, and the start and stop states of the electrical appliances may also collide. Also, the characteristic quantity is not obvious when the electrical appliance is running at low power (less than 100W). These have caused confusion and disappearance of load characteristic quantities, and led to a decrease in the identification accuracy of sub-item electrical appliances. These problems need a further study.

IV. Conclusion

In this paper, NILM scheme is introduced, and some key parts of NILM algorithm are studied. According to the concept of IR46, an intelligent electricity meter based on multicore module scheme is designed and developed. A multi-layer tree classification load matching algorithm is employed by the electric electricity meter to accurately identify the types of electrical appliances. The power composition accuracy test is carried out. The test results show that the power decomposition accuracy can reach 95-98% under single electrical appliance operation condition, and that is 70% -90% under multi-electrical hybrid operation. The research results are of some significance for analyzing the law of household electricity consumption, and carrying out precise energy use services.

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